

Electrochemistry

1. Electrolytes and non-electrolytes.
2. Strong electrolytes and weak electrolytes
3. Ionization and dissociation
4. Mobility of ions
5. Arrhenius theory of ionization.
6. Faraday's laws of electrolysis.
7. Transport number
8. Exhibition of electrolysis process in a cell.
9. Reversible cells & irreversible cells & molar conductance.
10. Mathematical problem.

✶ Electrolyte : An electrolyte is a compound that conducts electric current when it is in aqueous solution state or in melted state.

✳ Electrolytes-

- All ionic compounds are electrolytes
- When ionic compounds dissolve, they break apart into ions which are then able to conduct current.

✳ Non-electrolyte : A non-electrolyte is a compound that doesn't conduct electric current in either aqueous solution state or in molten state.

- Almost all covalent compounds aren't electrolytes (i.e., non-electrolytes).
- When these compounds dissolve in water, they don't produce ions.

✶ Difference between electrolytes and non-electrolytes:

Electrolytes	Non-electrolytes
1. Substances When dissolved in water, allow for conduct electricity are called electrolytes.	1. Substances When dissolved in water, doesn't allow for conduct electricity are called non-electrolytes.
2. They produce ions in water.	2. They don't produce ions in water.
3. These substances are formed by ionic bond.	3. These substances are formed by covalent bond.
4. Acids, bases and salts are electrolytes.	4. Sugar, honey etc. covalent compounds are non-electrolytes.

⇒ Strong electrolyte: A strong electrolyte is a solution that completely or almost completely ionizes or dissociates in a solution.

→ Strong electrolytes are good conductors of electricity.

⇒ The strong electrolytes may be strong acids, strong bases or soluble ionic salts.

→ The strong electrolytes are:

i. Strong acids: ^{eg.} HCl , H_2SO_4 , HNO_3 , HClO_4 , HBr and HI etc.

ii. Strong bases: NaOH , KOH , Ca(OH)_2 , Mg(OH)_2 etc.

iii. Salts: Practically all salts (NaCl , KCl) are strong electrolytes.

⇒ Weak electrolyte: A weak electrolyte is a solution that isn't completely ionizes or dissociates in a solution.

→ The weak electrolytes may be organic acids, organic bases or few salts that don't soluble in water.

→ The weak electrolytes are:

- i. Organic acids [e.g., CH_3COOH (acetic acid), Oxalic acid ($\text{HOOC}-\text{COOH}$), Sulphurous acid (H_2SO_3) etc.
- ii. Organic bases [e.g., $\text{C}_2\text{H}_5\text{NH}_2$ (Alkylamine)]
- iii. A few salts like Mercury(II) Chloride, Lead(II) acetate.

~~✗~~ Roles of electrolytes in the body:

- i. Regulate nerve and muscle function.
- ii. Balance the amount of water in the body.
- iii. Move wastes out from body.
- iv. Keep balance body's pH level.
- v. Help to rebuild damaged tissues.
- vi. Move nutrients into cells.
- vii. Keep blood pressure stable.
- viii. Keep heart rate and rhythm steady.

✚ Ionization: Ionization refers to the reaction in which the polar covalent compounds are converted into ions in water.

→ It is the process that involves the formation of ions.

→ It involves the creation of charges across the participating species.

✚ Dissociation: Dissociation refers to the separation of ions which are already present in electrovalent or ionic compounds.

→ It is the process of breaking up of a moiety (half) into its constituent atoms, molecules and ions.

→ It occurs due to a weak bond between species.

⇒ Difference between ionization and dissociation:

Ionization	Dissociation
1. It is the process which produces new charged particles.	1. It is the separation of charged particles which already exists in a compound.
2. It involves polar covalent compounds or metals.	2. It involves ionic compounds.
3. Irreversible.	3. Reversible
4. Involve covalent bond between atoms.	4. Involve ionic bond in compound.
5. Always produces charged particles.	5. It produces either charged particles or electrically neutral particles.

⇒ Conductance: Conductance is the ability of an element to conduct electric current. Conductance is expressed by G .

$$\text{So, } G = \frac{1}{R}.$$

Unit of conductance is Siemens (S).

☞ Mobility of ions / Relative speed of ions:

It is already considered that ions move to the oppositely charged electrodes under the influence of the electric current.

But the speeds of cations migrating towards the anode are not necessarily the same.

However, the speed of a cation moving away from the anode will be proportional to the fall of concentration of cations around the anode.

Similarly, the speed of anion moving away from the cathode will be proportional to the fall of concentration of anions around the cathode.

✱. Hittorf's gave a general rule is known as Hittorf's Rule.

It states that,

"The loss of concentration around any electrode is proportional to the speed of ions moving away from it."

→ Illustration of Hittorf's Rule:

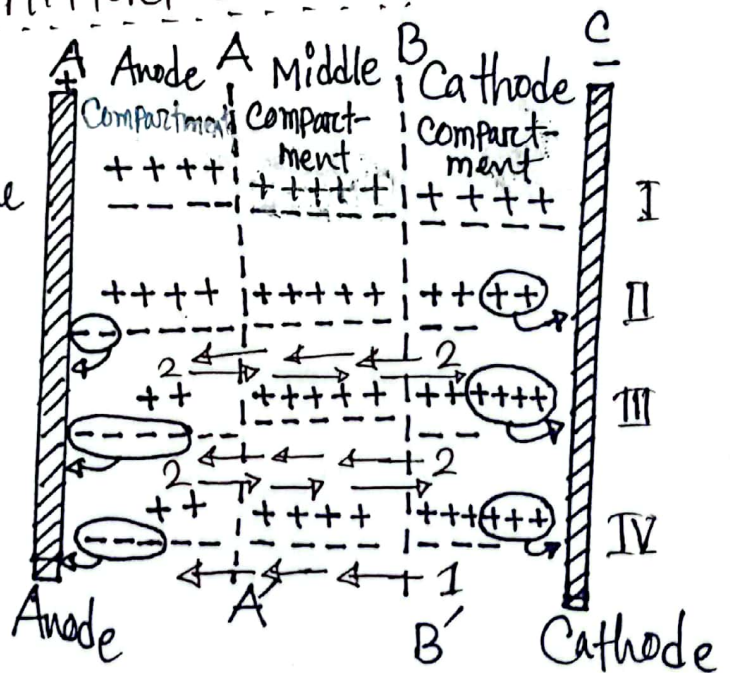
In the figure,

A is the anode. C is the cathode.

AA' and BB' are two imaginary planes which divide the cell into three compartments.

They are -

- i) Anode compartment
- ii) Middle compartment
- iii) Cathode compartment.



→ The sign (+) represents a cation while the sign (-) represents an anion.

Before electrolysis, let there be 13 ion-pairs in the cell. The number of ion-pairs in the two outer compartments is 4 each and there are 5 ion-pairs in the middle compartment (Position I).

Now let us consider the following cases.

(i) Let the anions alone be capable of movement:

When two anions have moved towards the anode, we get the position as shown in II. The cations have not moved at all.

The number of discharged anions and cations is the same viz., 2.

The concentration in the anode compartment, however, has not altered while in the cathode compartment it has fallen by two ion-pairs.

ii) Let the anions and cations move at the same rate:

When 2 ions of each type have crossed over towards the opposite electrodes, we get the condition shown in III. The number of discharged anions and cations is the same viz., 4. The concentration of both the anode compartment and the cathode compartment has fallen to the same extent viz, by two ion pairs.

iii) Let the cations move at twice the speed of the anions:

When cations have moved to the cathode compartment, one anion passes into the anode compartment. This state is shown in IV. The total number of discharged anions and cations is again same. Although the concentration in the cathode compartment has fallen by one ion-pair, anode compartment has fallen by two ion-pairs.

It is evident from the above considerations that ions are always discharged in equivalent amounts on the opposite electrodes.

We also conclude that the loss in concentration around any electrode is proportional to the speed of the ion moving away from it.

This is Hittorf's Rule. We can write the expression:

$$\frac{\text{Fall around anode}}{\text{Fall around cathode}} = \frac{\text{Speed of cation}}{\text{Speed of anion}} = \frac{V_+}{V_-}$$

Where, V_+ and V_- is the speed of cations and anions respectively.

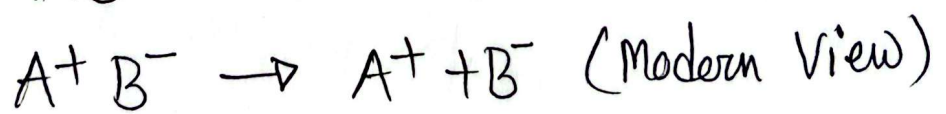
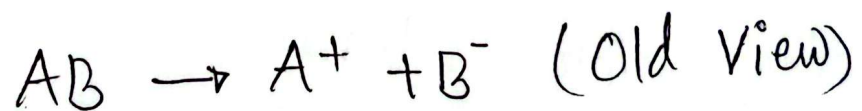
Ex. Arrhenius Theory of Ionization:

Savante Arrhenius studied the conduction of current through water solution of electrolytes. He came to believe that the conductivity of solutions was due to the presence of ions.

→ Arrhenius theory of ionisation may be stated as:

- (1) When dissolved in water, neutral electrolyte molecules are split up into two types of charged particles. These particles were called ions and the process was termed as ionisation. The positively charged were called cations and the negatively charged particles were called anions. In its modern form, the theory assumes that the ions are already present in the solid electrolyte and these are held together by electrostatic force. When placed in water, these neutral molecules dissociates to form separate cations & anions.

Thus,



For that reason, this theory may be referred to as theory of electrolytic dissociation.

- (2) The ions present in solution constantly reunite to form neutral molecules.

Thus there is a state of equilibrium between the dissociated molecules and the ions.



Applying the Law of Mass Action to the ionic equilibrium, we have,

$$K = \frac{[A^+][B^-]}{[AB]}$$

Where, K is called the Dissociation Constant.

Faraday's Law of Electrolysis:

Michael Faraday "described the qualitative aspects of electrolysis."

⇒ First Law of Faraday:

"The amount of substance deposited or liberated at an electrode during electrolysis is directly proportional to the quantity of charge (electricity) passes through the electrolyte solution."

If m is the mass of substance (in grams) deposited on electrode by passing Q Coulombs of electricity, then

$$m \propto Q$$

$$m \propto It \quad [\because Q = It]$$

$$m = ZIt$$

Where, 'Z' is the constant known as the electrochemical equivalent of the substance (electrolyte) and 'I' is the strength of current in amperes and 't' is the time in seconds for which the current has been passed.

If $I = 1$ ampere and $t = 1$ second, then

$$m = Z$$

Thus, the electrochemical equivalent is the amount of a substance deposited by 1 ampere current passing for 1 second (i.e., one Coulomb)

$$Z = \frac{\text{Equivalent mass of substance}}{\text{Faraday Constant}} = \frac{E}{F} = \frac{E}{96500}$$

$$\text{Faraday Constant, } F = 96500.$$

⇒ Ionic Mobility:

Ionic mobility is defined as the distance travelled by an ion per second under potential gradient of one volt per meter.

→ Potential Gradient:

Potential gradient is given by the potential difference ~~be~~ applied at the electrodes divided by distance between the electrodes.

$$\text{Ionic mobility} = \frac{\text{Speed of the ions in m s}^{-1}}{\text{Potential gradient in volt/m}}$$

Unit of ionic mobility : $\text{m}^2 \text{V}^{-1} \text{s}^{-1}$.

Faraday's 2nd Law of Electrolysis:

The mass of a substance deposited or liberated at any electrode on passing a certain amount of charge is directly proportional to its chemical equivalent weight. i.e., $m \propto E$.

$$\frac{m_1}{m_2} = \frac{Eq_1}{Eq_2}$$

Ex. Transport Number:

The fraction of total electric current carried by the cation or the anion is termed its transport number or Hittorf's number.

If V_+ represents the speed of migration of the cation and V_- that of anion.

$$\text{The transport number of cation} = \frac{V_+}{V_+ + V_-}$$

$$\text{The transport number of anion} = \frac{V_-}{V_+ + V_-}$$

The transport number of the cation is represented by t_+ and that of the anion by t_- .

$$t_+ = \frac{V_+}{V_+ + V_-} \text{ and } t_- = \frac{V_-}{V_+ + V_-}$$

$$\frac{t_+}{t_-} = \frac{V_+}{V_-} \text{ and } t_+ + t_- = 1$$

If the speed ratio V_+/V_- be denoted by π , we have

$$\pi = \frac{t_+}{t_-} = \frac{t_+}{1 - t_+}$$

$$t_- = \frac{1}{1 + \pi}$$

⇒ Application of Transport numbers or transference numbers:

1. Determining single-ion conductances:

In symmetric electrolytes, transport numbers can be used to precisely determine single-ion conductances.

2. Investigation of ion aggregates:

Transport numbers can be used to study ion aggregates, such as ion pairs in non-symmetric electrolytes and triple ions in symmetric electrolytes.

3. Estimating electrolytes:

Transport numbers ~~are~~ an important property to consider when estimating electrolytes because they provide an essential route for their ion transport.

4. Identifying battery function:

In batteries, the transport number of the primary ion type can be used to identify whether the battery can function normally and efficiently. For example, in Lithium-ion batteries (LIBs), the transport number of Li^+ (t_{Li^+}) affects the battery's charge-discharge rate, energy density and service life.

5. Maintaining constant electrolyte concentration:

In single-ion devices, such as lithium-ion batteries, electrolytes with a transport number close to 1 can maintain a constant electrolyte concentration during charge-discharge cycles.

"Transport numbers are the fraction of the total current carried by a given ion in an electrolyte."

☞ Molar Conductance:

It is defined as the conducting power of all the ions produced by 1 mole of an electrolyte in the given solution.

It is represented by μ .

$$\therefore \mu = K \times V$$

$$\mu = K \times 1000 / M$$

Where, V = Volume of solⁿ in cc (cm³).

K = Specific conduction of the solⁿ

M = Molarity of the solⁿ

☞ Electrolysis: The phenomenon of decomposition of an electrolyte by passing electric current through its solution is termed electrolysis.

☞ Electrolysis process in a cell:

→ Mechanism: How the electrolysis actually takes place, is illustrated in Fig. A. The cation migrate to the cathode and form a neutral

atom by accepting electrons from it. The anions migrate to the anode and yield a neutral particle by transfer of electrons to it. As a result of the loss of electrons by anions and gain of electrons by cations at their respective electrodes, chemical reaction takes place.

Example:

Let us consider electrolysis of HCl as an example. In solution, HCl is ionised,

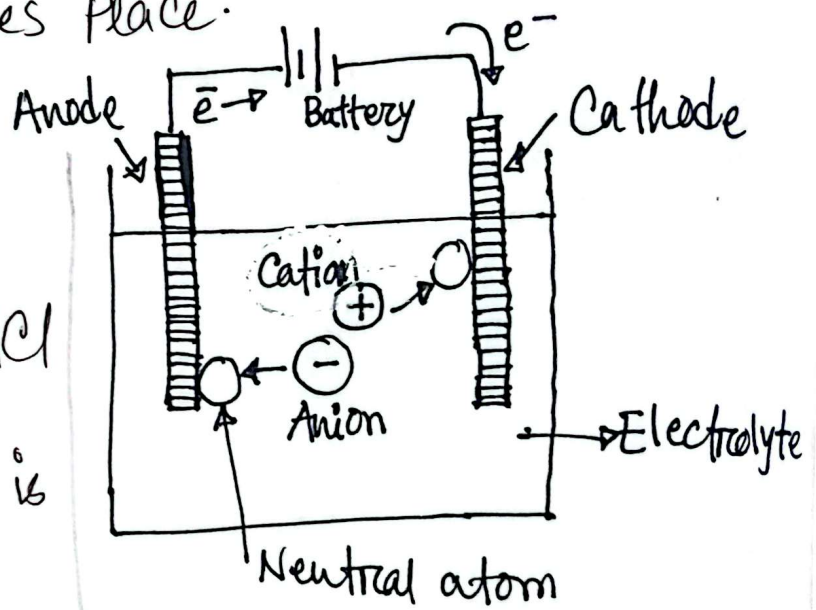
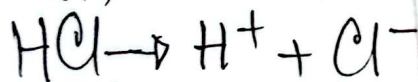
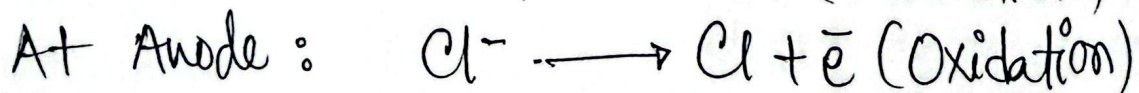
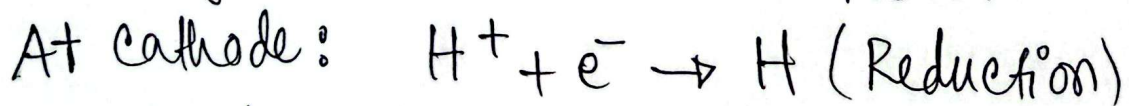
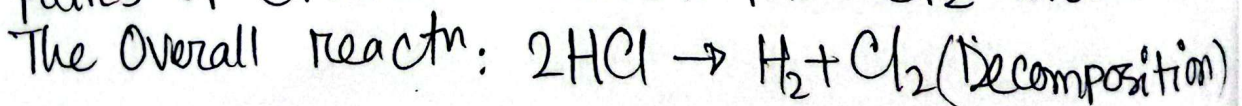


Fig- A: Mechanism of electrolysis

In the electrolytic cell Cl^- ions will move towards the anode and H^+ ions will move towards the Cathode. At the electrode, the following reaction will take place.



Pairs of H atom combines to form H_2 molecule and pairs of Cl atom combines to form Cl_2 molecule.



Mathematical Problems

[1.] The speed ratio of silver and nitrate ions in a solution of silver nitrate electrolysed between silver electrode is 0.916. Find the transport number of the two ion.

Solⁿ: We know that,

$$t_- = \frac{1}{1+r}$$

Where, t_- is the transport number of the anion and r is the speed ratio of the anion and the cation.

$$t_{\text{NO}_3^-} = \frac{1}{1+0.916} = 0.521$$

$$\begin{aligned} \text{And, } t_{\text{Ag}^+} &= 1 - t_{\text{NO}_3^-} = 1 - 0.521 \\ &= 0.479. \end{aligned}$$

[2.] A solution of silver nitrate containing 12.14 g of silver in 50 ml of solution was electrolysed between platinum electrodes. After electrolysis, 50 ml of the anode solution was found to contain 11.55 g of silver, while 1.25 g of metallic silver was deposited on the cathode. Calculate the transport number of Ag^+ and NO_3^- ions.

Solⁿ: Atomic weight of Ag = 108 g

Weight of Ag in 50 ml of the solⁿ before electrolysis = 12.14 g

Weight of Ag in 50 ml of the solⁿ after electrolysis = 11.55 g

Fall in concentration of Ag = $(12.14 - 11.55) \text{ g}$
 $= 0.59 \text{ g} = \frac{0.59}{108}$
 $= 0.0055 \text{ g eq.}$

Weight of Ag deposited in silver coulometer = 1.25 g
 $= \frac{1.25}{108} \text{ g eq.}$
 $= 0.0116$

Hence,

Transport number of Ag^+ (t_{Ag^+})

$$= \frac{\text{Fall in conc. around anode}}{\text{No. of g. eqvt deposited in silver coulometer}}$$

$$= \frac{0.0055}{0.0116}$$

$$= 0.474$$

$\therefore$ Transport number of NO_3^- ($t_{\text{NO}_3^-}$) -

$$= 1 - 0.474$$

$$= 0.526 \text{ (Ans.)}$$

[3.] In an electrolysis of copper sulphate between copper electrodes the total mass of copper deposited at the cathode was 0.153 g and the masses of copper per unit volume of the anode liquid before and after electrolysis were 0.79 and 0.91 g respectively. Calculate the transport number of the Cu^{2+} and SO_4^{2-} ions.

Solⁿ:

Wt. of copper in the anode liquid before
electrolysis = 0.79 g

Wt. of copper in the anode liquid after
electrolysis = 0.91 g

$$\begin{aligned}\text{Increase in weight} &= (0.91 - 0.79) \text{ g} \\ &= 0.12 \text{ g}\end{aligned}$$

Increase in weight of copper cathode
in the coulometer = 0.153 g (Given)

This means that if no copper had migrated
from the anode, increase in weight
would have been 0.153 g.

But actual increase = 0.12 g

Fall in conc. due to migration of Cu^{2+} ions

$$\begin{aligned}&= (0.153 - 0.12) \text{ g} \\ &= 0.033 \text{ g}\end{aligned}$$

$$\begin{aligned}\therefore \text{Transport number of } \text{Cu}^{2+} \text{ ion} &= \frac{0.033}{0.153} \\ &= 0.215\end{aligned}$$

$$\begin{aligned}\therefore \text{Transport number of } \text{SO}_4^{2-} \text{ ion} &= (1 - 0.215) \\ &= 0.785 \text{ (Ans)}\end{aligned}$$